DO NOW — RECALL

# Floating-point subtraction

Name: _____ Class: _____ Date: _____

**1. Why must floating-point exponents match before the mantissas are added directly?**

**2. Convert 01100000 1110 to denary. Use an eight-bit two’s-complement mantissa and a four-bit two’s-complement exponent.**

**3. Convert $6D_{16}$ to eight-bit binary.**

**4. Use 8-bit two’s complement to calculate $18 - 29$ . Show how subtraction becomes addition.**

DO NOW — CHECK / OBJECTIVES

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

### 1. Model answer

Because corresponding mantissa bit positions must represent the same scaled place values . Aligning the exponents puts both operands on the same scale before the mantissas are added.

### 2. Model answer

0.1875 Mantissa 01100000 = 0.75 ; exponent 1110 = −2 ; $0.75 \times 2^{-2} = 0.75 \div 4 = 0.1875$ .

### 3. Model answer

$01101101_2$ 6 → 0110 and D → 1101 .

### 4. Model answer

+18 = 00010010 +29 = 00011101 −29 = 11100011 00010010 + 11100011 = 11110101 11110101 represents −11 . So 18 − 29 = 18 + (−29) = −11 .

### By the end, you can…

• align exponents and subtract by adding a negated mantissa;

• handle positive, negative and zero differences;

• retain working precision and normalise after cancellation.

### Vocabulary — know these terms

Recall the term — Negation

Recall the term — Cancellation

Recall the term — Guard bits

Recall the term — Arithmetic right shift

Recall the term — Renormalisation

Recall the term — Underflow

NEW LEARNING

# Read, annotate and explain

**A − B = A + (−B)**

1. Align exponents
2. Keep guard bits
3. Negate B's mantissa
4. Add
5. Normalise

Negate a two’s-complement mantissa by inverting the working bits and adding 1.

Do not negate the exponent. It controls scale, not the sign of the operand.

### Align before subtracting

Choose a common exponent and use arithmetic right shifts to align the mantissas. Keep any low-order guard bits needed for an exact working result.

### Negate the second mantissa

At the chosen working width, invert every bit of B’s aligned mantissa and add 1.

### Add the aligned values

Sign-extend both operands to the same working width, add them, and interpret the signed result using the common exponent.

### Handle cancellation, zero and range

A small difference may need left shifts and a reduced exponent to become normalised. Equal operands produce zero.

**Explain one step in your own words, using the correct technical terms.**

I DO — WORKED EXAMPLE

# Follow the worked method

### Subtract 1.5 from 5

### Use common exponent 3

Align both mantissas using sign-preserving right shifts.

```
A: 0010100
B: 0000110
```

### Invert and add 1

```
B:        0000110
invert →  1111001
add 1  →  1111010
```

Negate the aligned mantissa, not the exponent.

### Calculate A + (−B)

```
0010100
 + 1111010
 = 0001110
```

The working difference is $14/32 \times 2^{3}$ .

### Shift left and reduce the exponent

```
0001110 × 2³
→ 011100 × 2²

Final: 011100 0010
```

Check: $5 - 1.5 = 3.5$ .





WE DO — CHECK AND IMPROVE

# Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

### Guided 1 — CHECK

Build the answer

### Guided 2 — CHECK

Build the answer

### Guided 3 — CHECK

Build the answer

### Guided 4 — CHECK

Build the answer
Recognise the all-zero result and stop; it does not need further normalisation.

### Guided 5 — CHECK

Build the answer

EXIT TICKET / REFLECTION

# Check your understanding

Complete the actual lesson Exit Ticket before checking.

**When subtracting B from A, only B's mantissa is negated before addition.**

**Circle: Confident / Getting there / Need help**

Explain what you can now do independently.

**GREEN PEN — Correct and explain**

Choose one error. Write the correct method and explain precisely why your first response was wrong.





WE DO — GUIDED PRACTICE

# Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

**Guided 1 — TRY: 2. Negative difference: 1.5 − 5**

Use a six-bit two’s-complement mantissa and a four-bit two’s-complement exponent. Give the final answer in normalised floating-point form before checking.

Method / working:

**Guided 2 — TRY: 3. Subtract a negative: 5 − (−1.5)**

Use a six-bit two’s-complement mantissa and a four-bit two’s-complement exponent. Give the final answer in normalised floating-point form before checking.

Method / working:

**Guided 3 — TRY: 4. Cancellation: 6.5 − 6**

Use a six-bit two’s-complement mantissa and a four-bit two’s-complement exponent. Give the final answer in normalised floating-point form before checking.

Method / working:

**Guided 4 — TRY: 5. Equal operands**

Use a six-bit two’s-complement mantissa and a four-bit two’s-complement exponent. Give the final answer in normalised floating-point form before checking.

Method / working:

**Guided 5 — TRY: OCR JUNE 2017 · Q6(d) · [6]**

Use a six-bit two’s-complement mantissa and a four-bit two’s-complement exponent. Give the final answer in normalised floating-point form before checking.

Method / working:

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

### Question 1

Question 1 · ADAPTED OCR EXAM PRACTICE · [6]
Calculate 010010 0010 minus 010010 0100 . Show exact alignment, negation, addition and an answer in normalised floating-point form using a six-bit two’s-complement mantissa and a four-bit two’s-complement exponent.
Adapted from OCR June 2017 Q6(d): operand order reversed; classroom marks.

### Question 2

Question 2 · ADAPTED EXAM PRACTICE · [5]
Calculate 010010 0100 minus 011010 0011 . Use a six-bit two’s-complement mantissa and a four-bit two’s-complement exponent. Show alignment and negation, then give the answer in normalised floating-point form.
Adapted from the OCR subtraction task: changed values; classroom marks.





YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

### Question 3

Question 3 · ORIGINAL EXAM-STYLE EXTENSION · [5]
Calculate 011000 0010 minus 100000 0001 using a six-bit two’s-complement mantissa and a four-bit two’s-complement exponent. Show how you handle the negative second operand and give the final answer in normalised floating-point form.

### Question 4

Question 4 · ORIGINAL EXAM-STYLE EXPLANATIONS · [4]
State the stored result when a value is subtracted from itself, using a six-bit two’s-complement mantissa and a four-bit two’s-complement exponent. [1]
Explain why normalising a small difference can require a lower exponent. [2]
Explain why a nonzero low-order bit must be retained during alignment where possible. [1]